

why 16 and not 4?

digitalisierung und programmierung · prof. dr. nicolas meseth

- 1 what your data says
- 2 what decides your alphabet
- 3 what it costs

part 1

what your data says

same light.
how far does it
get through?

image: ai-generated (gpt-image-2)

what your data says

your results

team	alphabet	hit rate	what they did about noise
team 7	16	98 %	cardboard tube, fixed distance, averaged over 8
team 2	8	96 %	longer integration time, shielded from the window
team 11	8	91 %	picked colours further apart
team 5	4	100 %	nothing, played it safe
team 14	4	82 %	nothing, and it still failed

same hardware. same task. a factor of four between first and last.

ten readings of one colour



the same colour
ten times

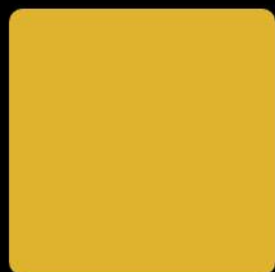
what the sensor reports

214

nothing changed. the numbers did.

what your data says

ten readings of one colour



the same colour
ten times

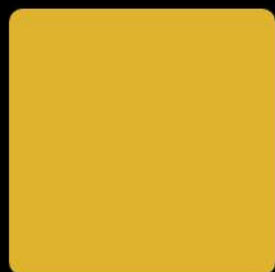
what the sensor reports

214 211 218 209 216 213 220 207 215 212

nothing changed. the numbers did.

what your data says

ten readings of one colour



the same colour
ten times

what the sensor reports

214 211 218 209 216 213 220 207 215 212



nothing changed. the numbers did.

what your data says

not the code. the scatter.

same sensor, same led, same room. the difference is how far your readings move when nothing moves.

part 2

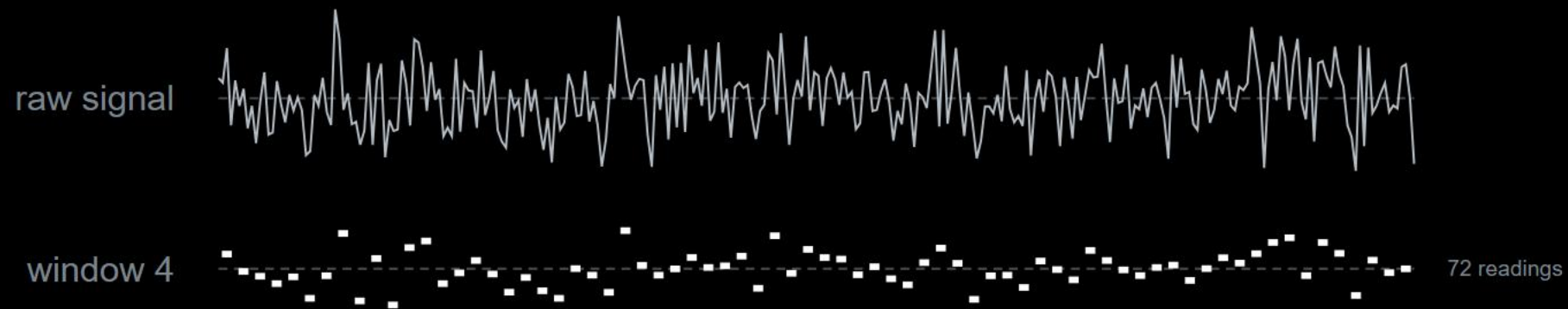
what decides your alphabet

the same signal, measured differently



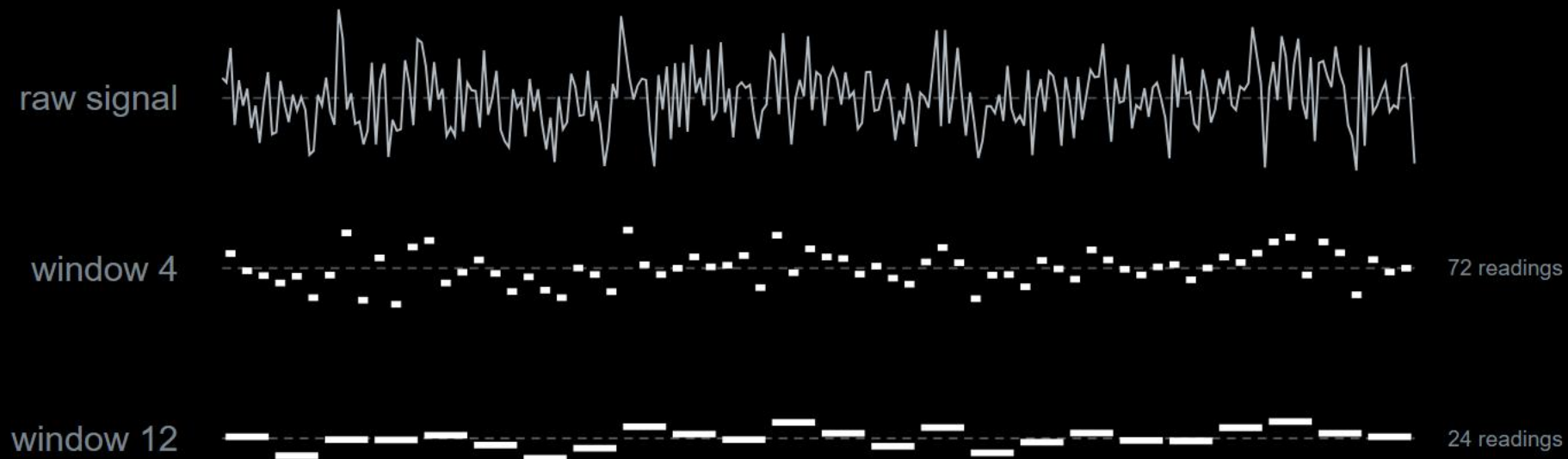
longer windows, calmer values. and fewer of them per second.

the same signal, measured differently



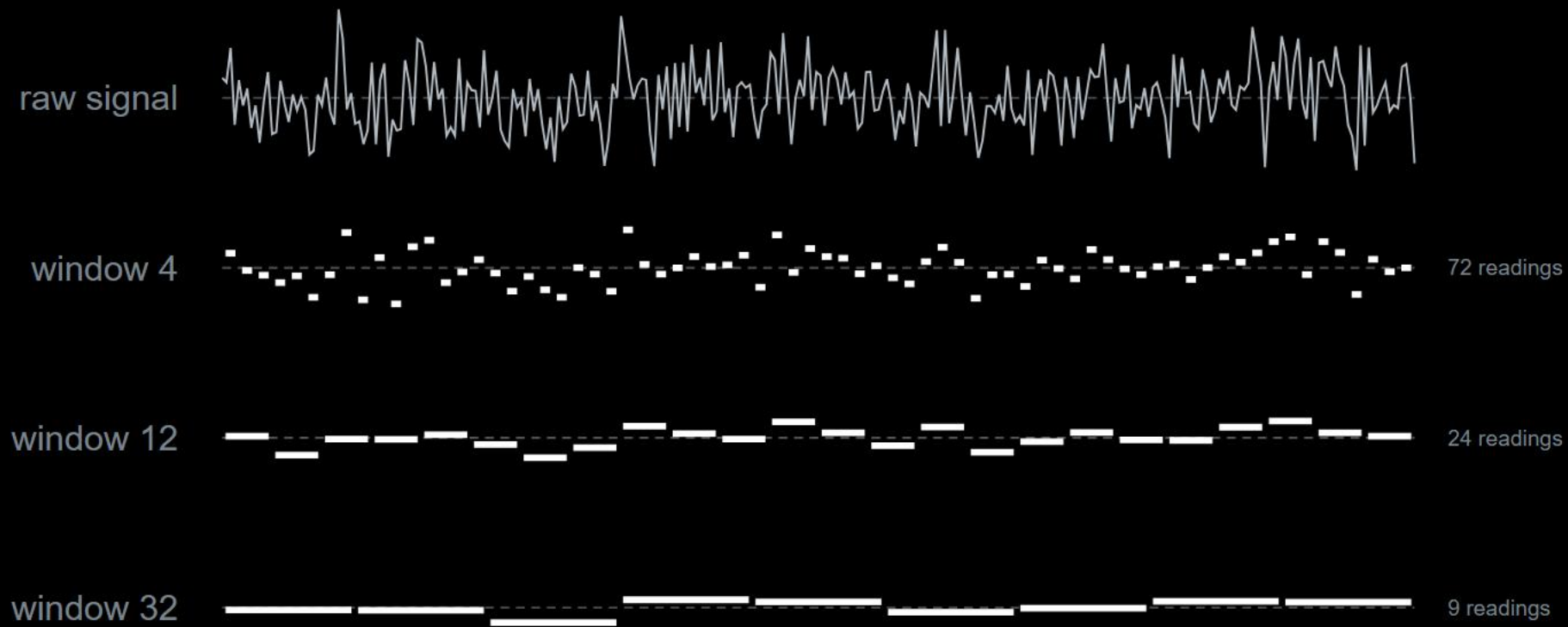
longer windows, calmer values. and fewer of them per second.

the same signal, measured differently



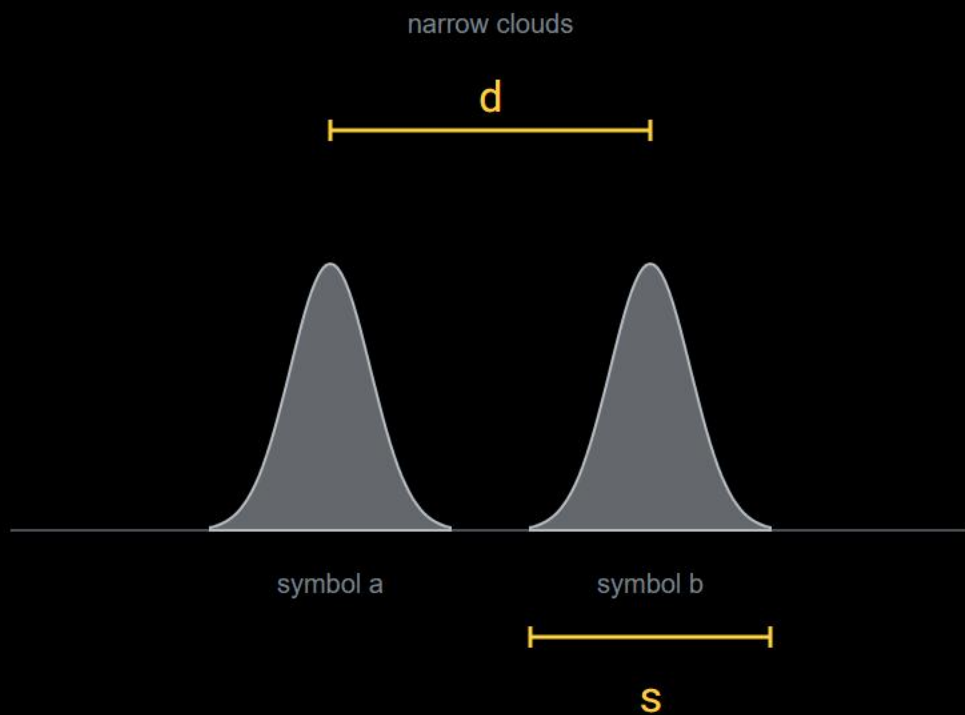
longer windows, calmer values. and fewer of them per second.

the same signal, measured differently



longer windows, calmer values. and fewer of them per second.

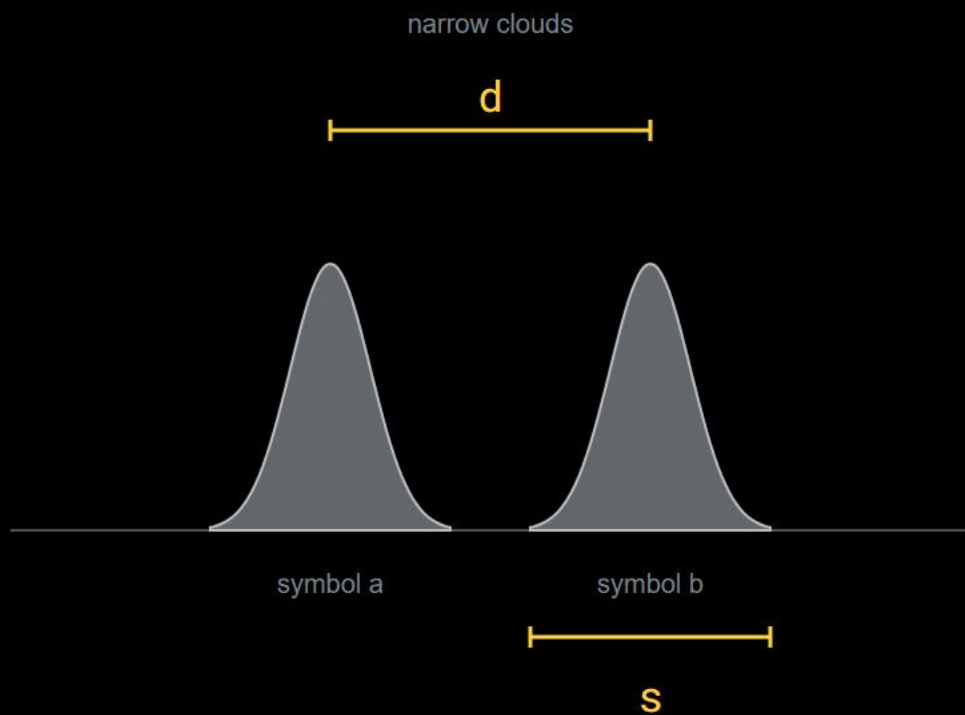
two symbols, two clouds



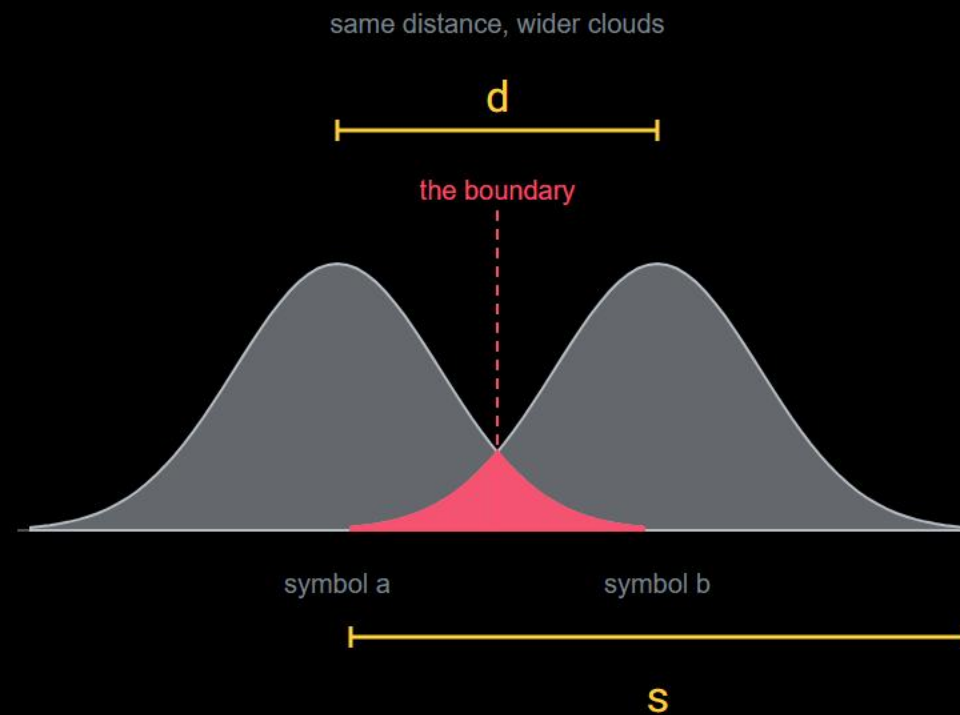
every reading lands on its own side

same distance. wider clouds. some readings land on the wrong side.

two symbols, two clouds



every reading lands on its own side



some readings land on the wrong side

same distance. wider clouds. some readings land on the wrong side.

the ratio is what counts

d / s

d distance between two symbols

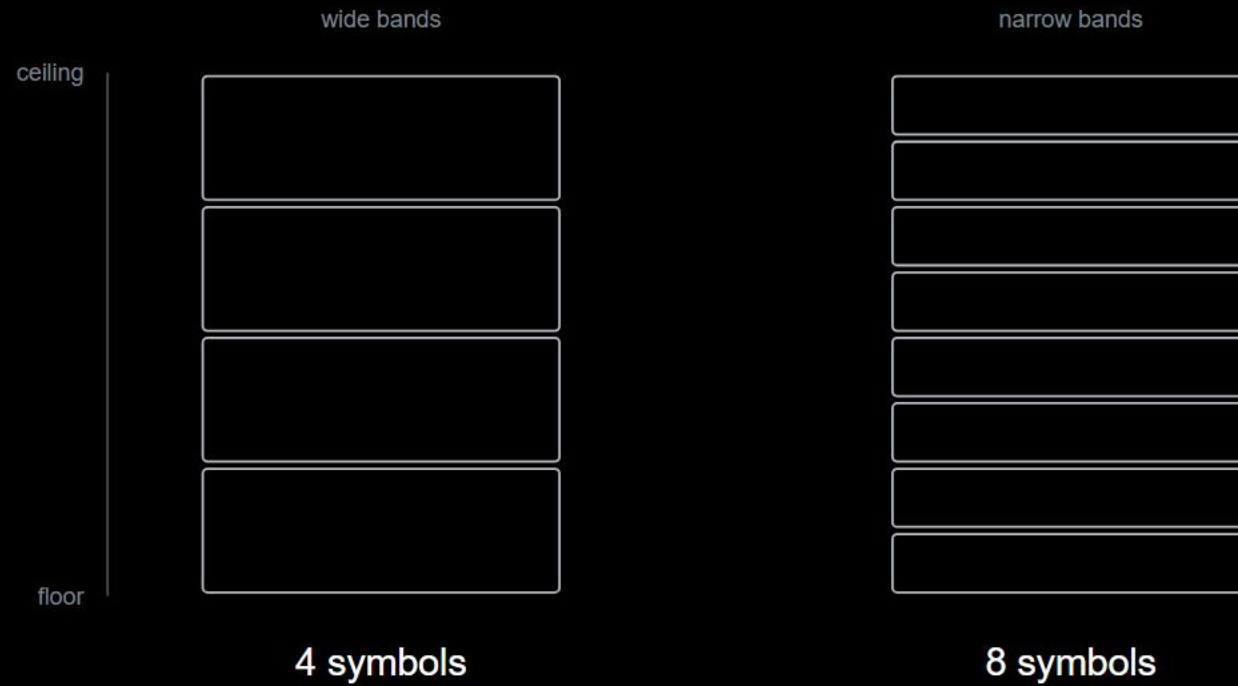
s spread of your readings



safely distinguishable means the clouds do not touch, not even at their edges

your alphabet size measures nothing but this ratio.

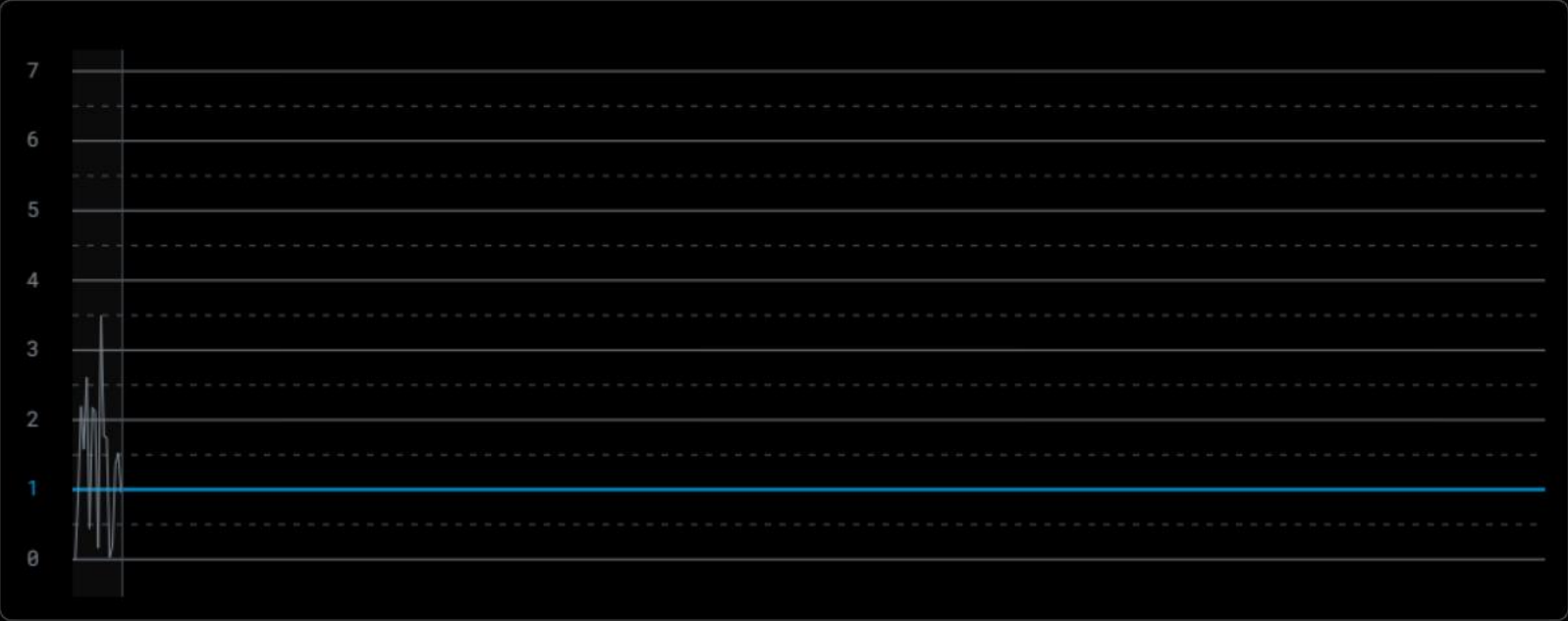
how many symbols fit?



the narrower your clouds, the more symbols fit into the same range

the range divided by the width of one band. that is your alphabet.

try it: the distinguishability lab



:received

simulated noise — the signal could be light, voltage or sound. the real numbers come from measuring your own link.

sending symbol

1

click a lane, or press its key

error rate

—

of the last 400 symbols

symbol rate

—

one symbol per window

bits per symbol

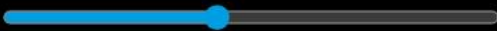
—

throughput if error-free

—

rate × bits per symbol

symbols in the alphabet



8 symbols (octal): 0 to 7

display speed



35 % of real time

measurement window (integration time)



120 ms: averages 24 raw readings

what decides your alphabet

part 3

what it costs

the toolbox, with price tags

bigger d

pick colours further apart

fewer symbols fit

use fewer symbols

fewer bits per symbol

aim the light better

only craft

smaller s

longer integration time

fewer symbols per second

average several readings

fewer symbols per second

block ambient light

only craft

fix the geometry

only craft

everything else is paid for in bits per symbol or symbols per second

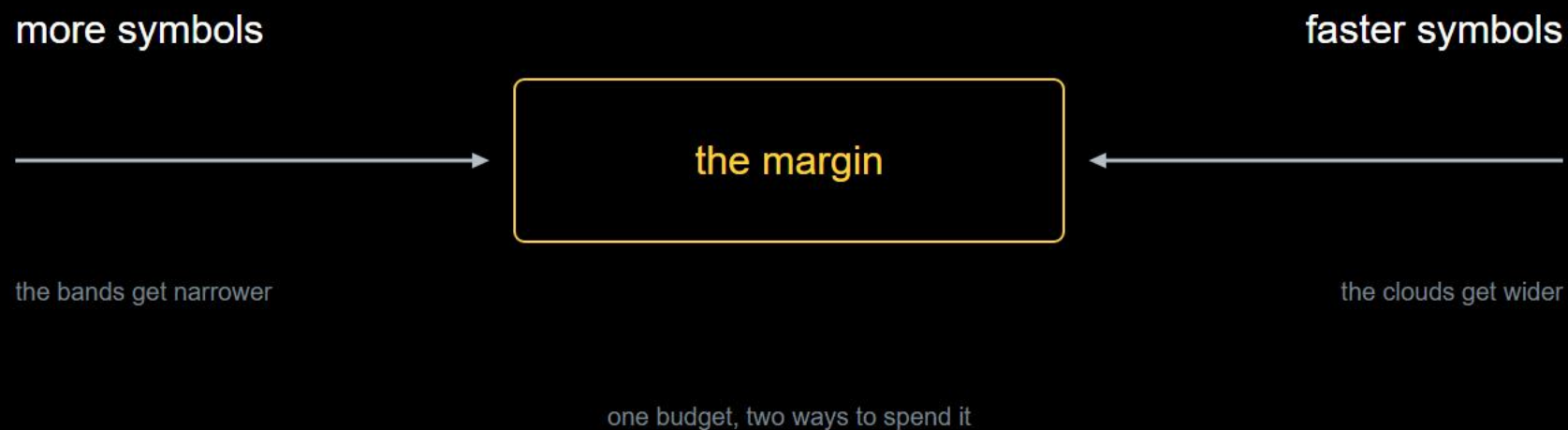
three of them cost nothing but craft. that is why the optics are allowed.

averaging is not free

```
def read_stable(n=10):  
    values = [sensor.get_clear() for _ in range(n)]  
    return sum(values) / n
```

ten times calmer readings. ten times slower.

both roads lead to the same wall



both spend the same margin. that wall is where challenge 2 begins.

and one honest sentence

we believed: more gain gives us more colours

we measured: the scatter grew with it, and we lost two symbols

the channel is the only authority in this room.

the fog stays.
measure how far
you see.

image: ai-generated (gpt-image-2)

what it costs